

FruitFit - Updated Product Requirements Document (PRD)

Vision

FruitFit is a browser-based fitness game inspired by Fruit Ninja where users perform fun arm workouts by slicing fruits in the air using their index finger. The front camera is enabled by default and becomes the game background.

Core Objectives

- Fun upper-body workout disguised as a game.
- Zero backend; everything runs locally.
- Camera-first experience with browser-based hand tracking.

Technology Stack

React 19, Vite, TypeScript, Tailwind CSS, HTML5 Canvas, MediaPipe Hand Landmarker (@mediapipe/tasks-vision), Zustand, Howler.js, Framer Motion, React Router, LocalStorage, optional IndexedDB, Vercel.

Gameplay Flow

Open app → Allow camera → Front camera starts → Countdown → Fruits spawn → User slices using fingertip → Score, combos and workout stats update → Statistics saved locally.

Critical Hand Tracking Requirements

- Highest priority feature.
- Target 60 FPS and perceived latency under 50 ms.
- Stable tracking during fast arm swings.
- Recover instantly if the hand leaves and re-enters view.
- Use all 21 hand landmarks for stability.
- Use the index fingertip as the virtual sword.
- Apply adaptive smoothing to reduce jitter while keeping controls responsive.
- Predict motion during rapid swings where appropriate.
- Slice detection must use the movement path between previous and current fingertip positions instead of a single point.
- Require minimum movement speed and distance before a slice is registered to avoid accidental cuts.
- Reward large arm swings over small wrist flicks to reinforce the workout goal.

Visual Feedback

Motion trail, juice splash, particles, combo effects, synchronized audio, subtle screen shake and responsive hit effects.

Workout Mode

3, 5, 10-minute and endless modes with estimated calories, arm swings, slices, combos and play history stored locally.

MVP

Front camera, camera switch, hand tracking, fruit slicing, bombs, combo system, workout timer, local statistics, responsive UI and sound effects.

Acceptance Criteria

Players should feel that the game follows their finger instantly, fast swings should register consistently, tracking should be smooth with minimal jitter, and gameplay should remain enjoyable on modern mobile browsers.